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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	MediAssist- AI Powered Hospital Appointment System	
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Abstract:

The **MediAssist- AI Powered Hospital Appointment System** Using SOA and Microservices is a web-based application designed to automate and simplify essential hospital activities such as patient registration, doctor management, appointment booking, billing, medical information management, and notifications. Traditional hospital management systems often depend on manual processes or tightly coupled applications, which can lead to delays, errors, increased workload, and difficulties in managing growing hospital operations. To overcome these challenges, the proposed system integrates **Artificial Intelligence, Agentic AI, Service-Oriented Architecture (SOA), and Microservices** to provide a secure, scalable, and intelligent hospital management solution.

The system consists of independent services such as **Patient Service, Doctor Service, Appointment Service, Billing Service, and Notification Service**. These services communicate through well-defined REST APIs and are managed using Spring Boot. Spring Cloud Eureka is used for service discovery, while Spring Cloud Gateway provides centralized routing and access to the services. The

system uses **Apache Kafka** for event-driven communication, allowing services to exchange events asynchronously and reducing direct dependency between services. **Spring Data JPA and MySQL** are used for data management, following the database-per-service approach.

To make the application intelligent and automated, the system incorporates **five AI-powered agents**: a **Patient Assistance Agent** for understanding patient queries, an **Appointment Agent** for assisting with doctor and appointment selection, a **Medical Information Agent** for summarizing patient-provided information, an **Emergency Triage Agent** for prioritizing potentially urgent cases, and a **Follow-up and Notification Agent** for managing reminders and follow-up activities. These agents interact with the SOA services to automate complete workflows instead of simply providing information to users. For example, when a patient requests an appointment, the Patient Assistance Agent understands the request, the Appointment Agent checks doctor availability, the Appointment Service confirms the booking, Kafka publishes an appointment event, and the Notification Agent sends a confirmation or reminder.

Security is provided using **Spring Security and JWT-based authentication**, with role-based access for patients, doctors, receptionists, and administrators. CORS is configured to control API access, while secure communication can be supported using TLS/SSL. The system also includes testing using JUnit 5, AssertJ, SpringBootTest, and DataJpaTest, while **GitHub Actions** is used to automate the build and testing process as part of the CI/CD workflow.

The major novelty of the proposed system is the integration of multiple cooperating AI agents with SOA-based hospital processes to provide intelligent automation. The system introduces AI-assisted emergency prioritization, intelligent appointment coordination, and automated patient follow-up as key innovative features. The AI components are designed as decision-support tools and do not replace professional medical diagnosis or clinical decision-making.

The project follows **Agile development practices**, where requirements are defined as user stories and organized into sprints using Jira. The proposed system supports **Sustainable Development Goal 3 (SDG 3): Good Health and Well-Being** by improving healthcare service coordination, reducing administrative workload, and providing faster and more accessible assistance to patients.

AI-DRIVEN HOSPITAL MANAGEMENT SYSTEM USING SOA AND MICROSERVICES

WORKFLOW DIAGRAM

